

Quality of Service

- **QoS Basics**

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- Functions (Scheduling, Metering, etc.)
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QoS Basics

1. What are the QoS attributes of a connection?

Bandwidth: The amount of data that can be transmitted per unit time.

Latency: The time delay between the transmission and reception of data packets.

Jitter: Variation in latency or delay between consecutive packets.

Packet Loss: Percentage of transmitted packets that are not received at the destination

Reliability: Measure of the consistency and predictability of network performance.

Prioritization: Assignment of priority levels to different types of traffic to ensure certain traffic receives preferential treatment.

2. What are the basic functions for the provisioning of QoS?

Traffic Classification: Identifying and categorizing different types of network traffic based on their QoS requirements

Traffic Policing: Enforcing traffic shaping and rate limiting policies to ensure compliance with QoS parameters.

Traffic Shaping: Regulating the flow of traffic to match specified QoS parameters, such as bandwidth allocation and latency requirements.

Congestion Management: Implementing mechanisms to manage network congestion and prioritize traffic based on QoS attributes.

3. Name the strategies for QoS provisioning.

Integrated Services (IntServ)

Differentiated Services (DiffServ)
Traffic Engineering
Quality of Experience (QoE) Management

4. Which applications have the toughest QoS requirements?

- Real-time multimedia applications such as video conferencing, VoIP (Voice over Internet Protocol), and live streaming.
- Online gaming and virtual reality applications that require low latency, minimal jitter, and high reliability for a smooth user experience.
- Financial trading platforms and high-frequency trading systems that demand low latency and minimal packet loss for real-time transaction processing.

5. Name measures to minimize the error rate over wireless links.

- Forward Error Correction (FEC): Adding redundant information to transmitted data to enable receivers to detect and correct errors without requiring retransmissions.
- Automatic Repeat reQuest (ARQ): Retransmitting data packets that are detected as corrupted or lost at the receiver to ensure reliable delivery

6. Name the measures to minimize the average delay over wireless links.

- Shorter Transmission Times: Using shorter transmission modes, such as higher modulation schemes or shorter packet sizes, to reduce the time taken to transmit data packets.
- Prioritized Queuing: Prioritizing time-sensitive traffic, such as voice or video packets, in the network queue to minimize queuing delays

7. Name the measures to maximize the throughput over wireless links.

- Channel Bonding: Aggregating multiple wireless channels to increase overall bandwidth and throughput.
- Dynamic Rate Adaptation: Adjusting transmission rates based on channel conditions to maximize data transfer rates while minimizing errors and interference.

8. How can you compute the maximal throughput of an end-to-end connection from its constituents?

The maximal throughput of an end-to-end connection can be computed by determining the bottleneck link in the network path and calculating the maximum achievable throughput of that link based on its capacity and utilization.

9. How compute the end-to-end delay for a connection for the case of zero transmission errors?

End-to-end delay for a connection with zero transmission errors can be computed by summing the propagation delay (time taken for a signal to travel from source to destination) and processing delay (time taken for packet processing at intermediate nodes) along the network path

10. What are the reasons of delay jitter, what the reasons for packet loss for wired and wireless communication?

- Delay Jitter: Variations in packet arrival times caused by fluctuations in network congestion, routing paths, and transmission delays.
- Packet Loss for Wired Communication: Packet loss in wired communication can be caused by network congestion, buffer overflow, hardware failures, or transmission errors.
- Packet Loss for Wireless Communication: Packet loss in wireless communication can be attributed to factors such as signal interference, fading, attenuation, and collisions in the wireless medium

QoS Functions and Common Concepts:

1. Which functions in a QoS-enabled network ensure that an application does not exceed its throughput requirements?

- Traffic Policing: Enforcing traffic rate limits to prevent an application from exceeding its allocated bandwidth.
- Traffic Shaping: Regulating the flow of traffic to ensure that the application's throughput stays within predefined limits
- Congestion Management: Prioritizing and managing traffic to prevent congestion and ensure sufficient bandwidth for critical applications.

2. How does the token bucket scheme helps in the limitation of the user throughput? How does the system support averaging?

The token bucket scheme limits user throughput by allowing packets to be transmitted only when tokens are available in the token bucket. Tokens are added to the bucket at a constant rate, representing the average allowed transmission rate. If the token bucket is full, excess tokens are discarded, limiting the user's throughput. The system supports averaging by maintaining a steady rate of token replenishment, ensuring that the average transmission rate matches the desired throughput.

3. What is the purpose of explicit packet dropping in routers?

Explicit packet dropping in routers is used to manage network congestion and ensure fair resource allocation. When congestion occurs, routers may drop packets selectively based on various criteria, such as packet priority, to alleviate congestion and maintain network performance.

4. How influences packet dropping the performance of TCP connections?

Packet dropping can trigger congestion control mechanisms in TCP, such as congestion avoidance and slow start. When TCP detects packet loss due to congestion, it reduces its congestion window, leading to lower throughput and potentially increased latency. Excessive packet dropping can significantly degrade the performance of TCP connections by causing frequent retransmissions and reduced throughput.

5. What are the ingredients of the TOS field? Why are multiple fields needed?

The TOS (Type of Service) field includes components such as precedence, delay, throughput, and reliability. Multiple fields are needed to provide granularity and flexibility in specifying QoS requirements for different types of traffic. Each field allows for the customization of QoS parameters based on the specific needs of applications and network services.

6. Why is the TOS field mostly relevant to the network as such rather than the end system?

The TOS field is mostly relevant to the network because it determines how routers and switches handle packets based on their QoS requirements. End systems may honor TOS markings when generating traffic, but the actual treatment of packets based on TOS markings occurs within the network infrastructure, where routers and switches prioritize and forward packets accordingly.

7. Why do networks differentiate between edge routers and core router? What's the purpose and what is the difference in behavior?

Networks differentiate between edge routers and core routers to optimize network performance, security, and resource utilization based on their respective roles within the network architecture.

Purpose:

Traffic Handling

QOS Enforcement

Security

Resource Allocation

The key difference in behavior between edge routers and core routers lies in their respective roles and responsibilities within the network architecture. Edge routers focus on traffic management, security, and service termination at network boundaries, while core routers prioritize efficient packet forwarding and routing within the internal network infrastructure.

QoS Strategies: IntServ, DiffServ, Overprovisioning:

1. What are the three main strategies for QoS provisioning in a network?

- Integrated Services (IntServ): Reservation-based approach that allocates network resources for specific traffic flows, ensuring end-to-end QoS guarantees.
- Differentiated Services (DiffServ): Class-based approach that prioritizes and treats traffic differently based on predefined service levels, allowing for scalable and flexible QoS provisioning
- Traffic Engineering: Optimization of network resources and routing paths to improve QoS parameters such as latency, bandwidth, and reliability

2. Why does IntServ not scale? What is the problem of packet handling per individual flow? Hint: think about the implementation of rate control for individual flows.

IntServ requires per-flow state and signaling for each individual flow, which can lead to scalability issues as the number of flows increases. The problem of packet handling per individual flow arises due to the need for maintaining state information and implementing rate control mechanisms for each flow, leading to increased overhead and complexity in router processing.

3. Why does RSVP use soft-states? What does it mean?

RSVP (Resource Reservation Protocol) uses soft-states to maintain signaling sessions and reservation states in routers. Soft-states have a finite lifetime and require periodic refreshment to remain active. If a soft-state is not refreshed within a specified timeout period, it is automatically removed. This approach helps prevent resource waste and ensures efficient resource allocation in the network.

4. How does RSVP handle the case that a router is not supporting it?

If a router does not support RSVP, RSVP-capable routers along the path can still establish end-to-end reservations by signalling with RSVP messages. Non-supporting routers simply forward RSVP messages without processing them and do not participate in reservation establishment. However, they must maintain packet forwarding and ensure that reserved traffic is not dropped or delayed.

5. What are the common entities in routers that are present in both IntServ and DiffServ systems?

- Traffic classifiers: Identifying and categorizing packets based on certain criteria such as source, destination, or type of service.
- Traffic conditioners: Enforcing traffic policies by shaping or policing traffic based on predefined rules.

- Queue management: Allocating bandwidth and managing packet queues to prioritize traffic based on its importance or QoS requirements.
- Packet scheduling: Determining the order in which packets are transmitted from the queue, considering QoS parameters

6. What are the major differences between IntServ and DiffServ wrt – Provided QoS requirements – Separation of flows – Effort for edge and core routers.

Provided QoS requirements: IntServ offers per-flow QoS, ensuring individual flows receive specific QoS guarantees, while DiffServ provides aggregate QoS, treating packets belonging to the same class uniformly.

Separation of flows: IntServ maintains per-flow state in routers, which can lead to scalability issues, whereas DiffServ relies on class-based packet marking and does not require per-flow state.

Effort for edge and core routers: IntServ requires more effort at both edge and core routers due to the need to maintain per-flow state and signaling, while DiffServ primarily involves edge routers for classification and marking, with minimal overhead in core routers.

7. How can we ensure that QoS works over multiple networks and multiple providers?

Standardization: Adhering to common QoS protocols and standards to ensure interoperability between different networks and providers.

Traffic engineering: Optimizing network resources and routing paths to prioritize QoS-sensitive traffic across multiple networks.

Service-level agreements (SLAs): Establishing agreements between network providers and customers regarding QoS guarantees and penalties for non-compliance.

Monitoring and management: Implementing tools and mechanisms to monitor QoS performance across network boundaries and take corrective actions as needed.

8. What is the difference between metering and marking?

Metering involves measuring the traffic characteristics, such as rate or size, to enforce traffic policies or QoS parameters.

Marking involves modifying packet headers or adding tags to indicate the desired treatment or priority level for the packet as it traverses the network.

9. Under what circumstances can overprovisioning be applied?

Traffic patterns are predictable: When the network experiences consistent and predictable traffic patterns, overprovisioning can ensure that there are sufficient resources to handle peak loads without degradation in performance.

Cost considerations: When the cost of additional network resources is lower than the potential losses from degraded performance or customer dissatisfaction due to congestion.

Uncertainty in demand: When it's challenging to accurately forecast future traffic demands, overprovisioning provides a buffer to accommodate unexpected spikes in traffic without impacting QoS.

QOS Signalling:

1. What's the idea behind NSIS?

The idea behind NSIS (Next Steps in Signaling) is to provide a framework for signaling in IP networks, allowing applications to request and negotiate Quality of Service (QoS) resources across the network. NSIS aims to improve the flexibility, scalability, and efficiency of signaling protocols compared to existing solutions like RSVP.

2. For which signaling purposes can NSIS be used?

Quality of Service (QoS) signaling: Requesting and negotiating QoS parameters such as bandwidth, delay, and packet loss.

Network resource reservation: Reserving network resources along a path to support specific application requirements.

Traffic engineering: Controlling and optimizing the routing of traffic flows based on network conditions and policies.

3. Does NSIS support secured signalling?

NSIS supports secured signaling through the use of security mechanisms such as Transport Layer Security (TLS) or IPsec. These mechanisms ensure the confidentiality, integrity, and authenticity of signaling messages exchanged between NSIS entities.

4. Does it use UDP or TCP for signaling?

NSIS typically uses UDP (User Datagram Protocol) for signaling. UDP is preferred over TCP (Transmission Control Protocol) due to its lower overhead and faster setup time, which is crucial for signaling protocols that require low latency and real-time responsiveness.

5. What is the purpose of the GIST sublayer?

The purpose of the GIST (General Internet Signaling Transport) sublayer in NSIS is to provide a reliable transport mechanism for NSIS signaling messages. GIST handles tasks such as

message fragmentation, reassembly, congestion control, and error recovery, ensuring the reliable delivery of signalling messages across the network.

QoS over Wireless:

1. Why is QoS considered a cross-layer issue?

QoS (Quality of Service) is considered a cross-layer issue because it involves coordination and optimization across multiple layers of the network stack, including the physical, data link, network, transport, and application layers. Different layers contribute to QoS in various ways, and optimizing QoS often requires considerations and adjustments at multiple layers simultaneously.

2. What influences the QoS of a wireless link? Can the physical layer adapt or influence the QoS? How? Are there limits of this?

The QoS of a wireless link is influenced by factors such as signal strength, interference, noise, modulation schemes, coding techniques, and channel conditions. The physical layer can adapt and influence QoS through techniques like adaptive modulation and coding, power control, beamforming, and channel bonding. However, there are limits to how much the physical layer can adapt, especially in highly dynamic and unpredictable wireless environments where factors like interference and fading can severely impact QoS.

3. Which PHY strategies are appropriate for the maximization of the throughput?

- High-order modulation: Using modulation schemes with more bits per symbol to increase data rate.
- Channel bonding: Aggregating multiple channels to increase bandwidth
- Spatial multiplexing: Transmitting multiple data streams simultaneously using multiple antennas (MIMO).

4. Which PHY strategies are appropriate for the minimization of the error rate?

- Forward Error Correction (FEC): Adding redundancy to transmitted data to enable error detection and correction.
- Adaptive coding and modulation: Dynamically adjusting coding and modulation schemes based on channel conditions to improve reliability.
- Diversity techniques: Using techniques such as space diversity, time diversity, or frequency diversity to mitigate the effects of fading and improve reliability

5. What are the drawbacks of the different approaches?

- Higher throughput strategies may be more susceptible to errors or interference
- Increasing redundancy to minimize error rate can decrease throughput efficiency
- Adaptive techniques may introduce complexity and overhead in implementation and operation

6. What is the benefit of ... or What would be a good strategy to optimize the system capacity of the cellular system as a whole? On which layer would this apply?

A good strategy to optimize the system capacity of a cellular system as a whole would be to implement techniques such as:

Spectrum efficiency improvements: Utilizing advanced modulation schemes, interference mitigation techniques, and spectrum sharing mechanisms to maximize the utilization of available frequency bands.

Network densification: Deploying more base stations to increase network coverage and capacity, especially in high-traffic areas.

HetNets (Heterogeneous Networks): Integrating diverse network elements like macrocells, small cells, and Wi-Fi hotspots to enhance coverage and capacity.

Carrier aggregation: Combining multiple frequency bands to increase bandwidth and data rates.

This optimization would primarily apply to the physical layer and also involve coordination with higher layers for efficient resource allocation and management.

QOS and Mobility:

1. What is the conflict between mobility management and QoS?

The conflict between mobility management and QoS arises from the inherent trade-off between maintaining seamless mobility for users and ensuring consistent QoS across changing network conditions. As users move between different network areas or access technologies, their QoS requirements may vary, and maintaining QoS guarantees while transitioning between network domains can be challenging.

2. What are possible solutions to this conflict?

Seamless handovers: Implementing efficient handover mechanisms that minimize disruption to ongoing sessions and ensure continuity of QoS parameters across different network domains.

Dynamic QoS adaptation: Adapting QoS parameters dynamically based on user mobility patterns, network conditions, and application requirements to maintain an acceptable level of service quality during handovers.

Cross-layer optimization: Coordinating mobility management and QoS mechanisms across multiple layers of the protocol stack to achieve better integration and alignment of QoS provisioning with mobility events.

3. What are the pros and cons of different approaches to the integration of QoS and mobility management?

The integration of QoS and mobility management can be approached in various ways, each with its own pros and cons:

Tight coupling: Integrating QoS and mobility management functions closely within the network architecture to provide seamless coordination and control. Pros include efficient resource utilization and better QoS support during mobility events, but cons may include increased complexity and overhead.

Loose coupling: Allowing QoS and mobility management functions to operate somewhat independently, with occasional coordination when necessary. Pros include flexibility and scalability, but cons may include potential delays or disruptions during handovers.

Cross-layer optimization: Optimizing QoS provisioning and mobility management across multiple layers of the protocol stack to achieve synergies and trade-offs. Pros include holistic management of QoS and mobility, but cons may include compatibility issues and interoperability challenges.